Unit 2: Unit II: Calculus (10 Hours)

Subject: Mathematics-I (BMAT-001) • **Exam Weightage:** 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Study Time:** 35 mins

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1. Mean Value Theorems & Power Series Expansions

1. In-Depth Analysis of the Three Fundamental Mean Value Theorems

Differential calculus is built on three interconnected foundation stones: Rolle's Theorem, Lagrange's Mean Value Theorem, and Cauchy's Generalized Mean Value Theorem.

A. Rolle's Theorem (1691)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be a real function satisfying:

1. f is **continuous** on the closed interval $[a, b]$.

2. f is **differentiable** on the open interval (a, b).

3. $f(a) = f(b)$.

> Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = 0}$$

- **Geometric Interpretation:** If a continuous smooth curve starts and ends at the same horizontal height, there must be at least one crest or trough where the tangent line is perfectly horizontal (slope = 0).

- **Physical Interpretation:** If a car returns to its starting point after a road trip, its instantaneous velocity must have been zero at least once when turning around.

B. Lagrange's Mean Value Theorem (First MVT)

✅ **FUNDAMENTAL LAW / RULE:**

Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = \frac{f(b) - f(a)}{b - a}}$$

- **Geometric Interpretation:** There exists at least one interior point c where the instantaneous tangent line is strictly parallel to the chord (secant line) joining the endpoints $(a, f(a))$ and $(b, f(b))$.
- **Alternative Form:** Setting $b = a + h$ and $c = a + \theta h$ with $0 < \theta < 1$:

$$f(a + h) = f(a) + hf'(a + \theta h)$$

C. Cauchy's Generalized Mean Value Theorem

✅ **FUNDAMENTAL LAW / RULE:**

Theorem Statement: Let two functions $f, g : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$, differentiable on (a, b), and $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists at least one $c \in (a, b)$ such that:

$$\mathbf{\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}}$$

- **Proof Architecture:** Define auxiliary function $\phi(x) = f(x) - \lambda g(x)$. Choose $\lambda = \frac{f(b)-f(a)}{g(b)-g(a)}$ so that $\phi(a) = \phi(b)$. Apply Rolle's theorem to $\phi(x)$ to obtain $\phi'(c) = f'(c) - \lambda g'(c) = 0$.

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2. Taylor's Theorem with Remainder & Power Series

Taylor's Theorem with Remainder Terms:

If $f(x)$ and its first $(n - 1)$ derivatives are continuous on $[a, a + h]$ and $f^{(n)}(x)$ exists on $(a, a + h)$, then:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

Where the remainder R_n can be expressed as:

- **Lagrange's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{n!}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$
- **Cauchy's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{(n-1)!}(1-\theta)^{n-1}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$

Maclaurin's Series Expansion (Centering at $a = 0$):

When $\lim_{n\to\infty} R_n = 0$, the function can be expressed as an infinite power series:

$$\mathbf{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots}$$

Standard Expansion Catalog for Exams (Expansion by Use of Known Series):

1. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots \quad (\forall x \in \mathrm{R})$
2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \quad (\forall x \in \mathrm{R})$
3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \quad (\forall x \in \mathrm{R})$
4. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \quad (-1 < x \leq 1)$
5. $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \ldots \quad (|x| < 1)$

Problem:
Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

Step 1 🚀: Check Rolle's Three Hypotheses

1. $f(x)$ is a polynomial, hence continuous on $[1, 3]$.
2. $f'(x) = 2x - 4$ exists on $(1, 3)$, hence differentiable on $(1, 3)$.
3. Check boundary values:

$$f(1) = 1^2 - 4(1) + 3 = 0$$

$$f(3) = 3^2 - 4(3) + 3 = 9 - 12 + 3 = 0 \implies f(1) = f(3)$$

Step 2 ⚡: Find Stationary Point $c \in (1, 3)$ **where** $f'(c) = 0$

$$f'(c) = 2c - 4 = 0 \implies \mathbf{c = 2}$$

Since $c = 2 \in (1, 3)$, Rolle's Theorem is **completely verified**!

🎯 Rolle's Theorem verified: $\mathbf{c = 2 \in (1, 3)}$.

University Question (4 Marks - MMDU Question Paper):
Expand $f(x) = e^x \cos x$ up to the term containing x^4 using Maclaurin's series (and verify by multiplication of known series).

Step 1 🚀: Compute Successive Derivatives at $x = 0$

- $f(x) = e^x \cos x \implies f(0) = 1$
- $f'(x) = e^x(\cos x - \sin x) \implies f'(0) = 1(1 - 0) = 1$
- $f''(x) = -2e^x \sin x \implies f''(0) = 0$
- $f'''(x) = -2e^x(\sin x + \cos x) \implies f'''(0) = -2(0 + 1) = -2$
- $f^{(4)}(x) = -4e^x \cos x \implies f^{(4)}(0) = -4(1) = -4$

Step 2 ⚡: Substitute into Maclaurin's Formula (or Known Series)

$$e^x \cos x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots) = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots$$

🎯 Maclaurin Expansion: $\mathbf{e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + O(x^5)}$.

2. Curvature, Centre & Chord of Curvature (Cartesian, Parametric, Polar)

1. Curvature, Radius & Center of Curvature: Comprehensive Theory

Physical Concept of Curvature (κ):

✅ **FUNDAMENTAL LAW / RULE:**

Definition: Let P and Q be two neighboring points on a smooth curve separated by arc length Δs. If the tangent vectors at P and Q make angles ψ and $\psi + \Delta\psi$ with the positive x-axis, the **Curvature** κ is the rate of turning of the tangent line per unit arc length:

$$\kappa = \lim_{\Delta s \to 0} \left| \frac{\Delta \psi}{\Delta s} \right| = \left| \frac{d\psi}{ds} \right|$$

Radius of Curvature (ρ):

The Radius of Curvature ρ is the reciprocal of the curvature:

$$\rho = \frac{1}{\kappa} = \left| \frac{ds}{d\psi} \right|$$

- A straight line has zero bending $\implies \kappa = 0 \implies \rho = \infty$.
- A circle of radius R bends uniformly $\implies \kappa = 1/R \implies \rho = R$.

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2. Analytical Formulas for Radius of Curvature ρ

Coordinate Representation	Curve Equation	Radius of Curvature Formula ρ
Cartesian $y = f(x)$	$y = f(x)$	$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$ $(y_1 = \frac{dy}{dx},\ y_2 = \frac{d^2y}{dx^2})$
Cartesian $x = g(y)$	$x = g(y)$	$\rho = \frac{(1+x_1^2)^{3/2}}{x_2}$ $(x_1 = \frac{dx}{dy},\ x_2 = \frac{d^2x}{dy^2})$
Parametric	$x = x(t),\ y = y(t)$	$\rho = \frac{(x'^2+y'^2)^{3/2}}{x'y''-y'x''}$
Polar Form	$r = f(\theta)$	$\rho = \frac{(r^2+r_1^2)^{3/2}}{r^2+2r_1^2-rr_2}$ $(r_1 = \frac{dr}{d\theta},\ r_2 = \frac{d^2r}{d\theta^2})$
Pedal Form (p-r)	$p = f(r)$	$\rho = r\frac{dr}{dp}$

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3. Center of Curvature $(\bar{x}, \bar{y})$ and Chord of Curvature

The circle that shares the same tangent and curvature as the curve at point $P(x, y)$ is the **Osculating Circle (Circle of Curvature)**. Its center $C(\bar{x}, \bar{y})$ is given by:

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}, \quad \bar{y} = y + \frac{1+y_1^2}{y_2}$$

- **Equation of Circle of Curvature:** $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$.
- **Evolute:** The locus of the centers of curvature of all points on a curve is called the **Evolute** of that curve.

Chords of Curvature:

- **Chord parallel to x-axis:** $C_x = 2\rho\sin\psi = \frac{2y_1(1+y_1^2)}{y_2}$
- **Chord parallel to y-axis:** $C_y = 2\rho\cos\psi = \frac{2(1+y_1^2)}{y_2}$
- **Chord through the Pole (Polar Curve $r = f(\theta)$):** $C_p = 2\rho\sin\phi = \frac{2r(r^2+r_1^2)}{r^2+2r_1^2-rr_2}$

University Question (6 Marks):
Find the radius of curvature at any point t on the cycloid $x = a(t+\sin t),\ y = a(1-\cos t)$.

Step 1 🚀: Compute First and Second Derivatives w.r.t t

- $x' = \frac{dx}{dt} = a(1+\cos t) = 2a\cos^2(\frac{t}{2})$

- $y' = \frac{dy}{dt} = a\sin t = 2a\sin(\frac{t}{2})\cos(\frac{t}{2})$

- $x'' = -a\sin t, \quad y'' = a\cos t$

Step 2 ⚡ : Denominator $x'y'' - y'x''$ and Numerator $(x'^2 + y'^2)^{3/2}$

$$x'y'' - y'x'' = a^2(1+\cos t) = 2a^2\cos^2(\frac{t}{2})$$

$$(x'^2 + y'^2)^{3/2} = [4a^2\cos^2(\frac{t}{2})]^{3/2} = 8a^3\cos^3(\frac{t}{2})$$

$$\rho = \frac{8a^3\cos^3(t/2)}{2a^2\cos^2(t/2)} = 4a\cos(\frac{t}{2})$$

🎯 Radius of Curvature: $\rho = \mathbf{4a}\cos(\frac{\mathbf{t}}{\mathbf{2}})$.

University Question (6 Marks - MMDU Question Paper):
Find the coordinates of the centre of curvature $(\bar{X}, \bar{Y})$ and the equation of the evolute of the parabola $y^2 = 4ax$.

Step 1 🚀 : Derivatives for $y^2 = 4ax$

$$2yy_1 = 4a \implies y_1 = \frac{2a}{y}, \quad y_2 = -\frac{4a^2}{y^3}$$

Step 2 ⚡ : Centre of Curvature Formula

$$\bar{X} = x - \frac{y_1(1+y_1^2)}{y_2} = x + \frac{y^2 + 4a^2}{2a} = 3x + 2a$$

$$\bar{Y} = y + \frac{1+y_1^2}{y_2} = -\frac{y^3}{4a^2}$$

Step 3 🎯: Eliminate (x, y) to Obtain the Evolute Equation
Substitute $x = \frac{\bar{X}-2a}{3}$ and $y^2 = (4a^2\bar{Y})^{2/3}$ into $y^2 = 4ax$:

$$27a\bar{Y}^2 = 4(\bar{X} - 2a)^3$$

🎯 Centre of Curvature: $(\bar{X}, \bar{Y}) = (3x + 2a, -\frac{y^3}{4a^2})$ | Evolute: $\mathbf{27aY^2 = 4(X - 2a)^3}$.

3. Indeterminate Forms & L'Hospital's Rule

1. Rigorous Analysis of Indeterminate Forms & L'Hospital's Rule

What is an Indeterminate Form?

An algebraic limit expression $\lim_{x\to a} \frac{f(x)}{g(x)}$ is called **indeterminate** if substituting the limiting point $x = a$ fails to yield a well-defined value because the limiting behaviors of the numerator and denominator conflict.

The 7 Classic Indeterminate Forms:

$$\frac{\mathbf{0}}{\mathbf{0}}, \quad \frac{\infty}{\infty}, \quad \mathbf{0 \times \infty}, \quad \infty - \infty, \quad \mathbf{0^0}, \quad \mathbf{1^\infty}, \quad \infty^{\mathbf{0}}$$

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2. L'Hospital's Rule Theorem & Derivation

✅ **FUNDAMENTAL LAW / RULE:**
Theorem (Bernoulli & L'Hospital): Let f and g be differentiable on an open interval containing a (except possibly at a). If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ (or both tend to $\pm\infty$), and $g'(x) \neq 0$ near a, then:

$$\lim_{\mathbf{x\to a}} \frac{\mathbf{f(x)}}{\mathbf{g(x)}} = \lim_{\mathbf{x\to a}} \frac{\mathbf{f'(x)}}{\mathbf{g'(x)}}$$

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3. Systematic Reduction Strategies for Non-Fractional Forms

1. **Form $0 \times \infty$:** Invert one factor into the denominator: $f(x) \cdot g(x) = \frac{f(x)}{1/g(x)} \implies \frac{0}{0}$.

2. **Form $\infty - \infty$:** Combine over a common denominator using LCM or rationalization.

3. **Exponential Forms ($0^0, 1^\infty, \infty^0$):** Let $y = [f(x)]^{g(x)}$, take $\ln y = g(x)\ln[f(x)]$, evaluate $L = \lim_{x\to a} \ln y$, and recover $\lim y = e^L$.

University Question (4 Marks):
Evaluate $\lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right)$.

Step 1 🚀: Convert $\infty - \infty$ into $\frac{0}{0}$ via LCM

$$\lim_{x\to 0}\left(\frac{\sin^2 x - x^2}{x^2 \sin^2 x}\right) = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4} \cdot \lim_{x\to 0}\left(\frac{x}{\sin x}\right)^2 = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4}$$

Step 2 ⚡: Use Series Expansion $\sin x = x - \frac{x^3}{6} + O(x^5)$

$$\sin^2 x - x^2 = \left(x - \frac{x^3}{6}\right)^2 - x^2 = -\frac{x^4}{3} + O(x^6)$$

$$\lim_{x\to 0} \frac{-\frac{x^4}{3} + O(x^6)}{x^4} = -\frac{1}{3}$$

🎯 Evaluated Limit: $-\frac{1}{3}$.

University Question (6 Marks - MMDU Question Paper):
Evaluate $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2}$.

Step 1 🚀: Identify 1^∞ Form & Take Natural Logarithm
Let $y = \lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2} \implies \ln y = \lim_{x\to 0} \frac{\ln\left(\frac{\tan x}{x}\right)}{x^2}$.

Step 2 ⚡: Series Expansion of $\tan x$

$$\frac{\tan x}{x} = 1 + \frac{x^2}{3} + O(x^4) \implies \ln\left(\frac{\tan x}{x}\right) = \frac{x^2}{3} + O(x^4)$$

Step 3 🎯: Evaluate Limit and Exponentiate

$$\ln y = \lim_{x\to 0} \frac{\frac{x^2}{3} + O(x^4)}{x^2} = \frac{1}{3} \implies y = e^{1/3} = \sqrt[3]{e}$$

🎯 Limit: $\mathbf{e^{1/3} = \sqrt[3]{e}}$.

4. Beta and Gamma Functions & Their Properties

1. Rigorous Theory of Beta and Gamma Special Functions

A. The Gamma Function (Euler's Integral of Second Kind):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For any real number $n > 0$, the **Gamma Function** $\Gamma(n)$ is defined as:

$$\boldsymbol{\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\, dx \quad (n > 0)}$$

Fundamental Properties of $\Gamma(n)$:

1. **Recurrence Relation:** $\boldsymbol{\Gamma(n+1) = n\,\Gamma(n)}$ and $\boldsymbol{\Gamma(n+1) = n!}$ for positive integers.
2. **Base & Half-Integer Values:** $\Gamma(1) = 1$ and $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$.

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2. The Beta Function (Euler's Integral of First Kind)

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For $m > 0, n > 0$, the **Beta Function** $B(m, n)$ is defined as:

$$\boldsymbol{B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx}$$

Key Properties and Master Identities:

1. **Symmetry:** $\mathbf{B(m, n) = B(n, m)}$.

2. **Trigonometric Form:**

$$\mathbf{B(m, n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta}$$

3. **Infinite Domain Form:**

$$\mathbf{B(m, n) = \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} \, dt}$$

4. **Beta-Gamma Bridge Relation:**

$$\mathbf{B(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}}$$

5. **Walli's / Trigonometric Integral Formula:**

$$\int_0^{\pi/2} \sin^{p}\theta \cos^{q}\theta \, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$$

6. **Legendre's Duplication Formula:**

$$\Gamma(m)\,\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}}\,\Gamma(2m)$$

University Question (6 Marks - MMDU Question Paper):
Evaluate $\int_0^{\pi/2} \sin^4\theta \cos^6\theta \, d\theta$ using Beta and Gamma functions, and prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$.

Step 1 🚀: Evaluate Trigonometric Integral ($p = 4, q = 6$)

$$I_1 = \frac{\Gamma(\frac{5}{2})\Gamma(\frac{7}{2})}{2\,\Gamma(6)} = \frac{(\frac{3\sqrt{\pi}}{4})(\frac{15\sqrt{\pi}}{8})}{2 \times 120} = \frac{3\pi}{512}$$

Step 2 ⚡: Prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$
Substitute $x^4 = u \implies x = u^{1/4},\ dx = \frac{1}{4}u^{-3/4}du$:

$$I_2 = \frac{1}{4}\int_0^1 u^{1/4-1}(1-u)^{1/2-1}du = \frac{1}{4}B(\frac{1}{4},\frac{1}{2}) = \frac{\Gamma(1/4)\Gamma(1/2)}{4\,\Gamma(3/4)} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$$

🎯 Integral Value: $\frac{3\pi}{512}$ | Formula proved via $B(\frac{1}{4},\frac{1}{2})$.

University Solved Examination Questions (Unit 2)

MMDU Official University Solved Examination Questions (Unit 2)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 **Question 4 (1 Mark): Geometric Interpretation of Rolle's Theorem**

Question:
State the geometric meaning of the point c in Rolle's Theorem.

Step-by-Step Solution:
If a continuous, differentiable curve $y = f(x)$ has equal values at the endpoints $f(a) = f(b)$, there exists at least one interior point $c \in (a, b)$ where $f'(c) = 0$, meaning the **tangent line at** $(c, f(c))$ **is parallel to the** x**-axis**.

🎯 **Verified Result:** Horizontal tangent parallel to x-axis at $x = c$.

📘 Question 14 (2 Marks): Radius of Curvature of Catenary Curve

Question:

Find the radius of curvature ρ for the catenary $y = c\cosh(x/c)$.

Step-by-Step Solution:

Given $y = c\cosh(x/c)$, we have $y_1 = \sinh(x/c)$ and $y_2 = \frac{1}{c}\cosh(x/c)$.
Since $1 + y_1^2 = 1 + \sinh^2(x/c) = \cosh^2(x/c)$:

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\cosh^3(x/c)}{\frac{1}{c}\cosh(x/c)} = c\cosh^2(x/c) = \frac{y^2}{c}$$

🎯 **Verified Result:** $\rho = \frac{y^2}{c}$.

📘 Question 22 (6 Marks): Relation Between Beta and Gamma Functions

Question:

State and prove $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and hence evaluate $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$.

Step-by-Step Solution:

Using polar coordinates $x = r\cos\theta, y = r\sin\theta$ in $\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1}\, dx\, dy$ yields $\Gamma(m)\Gamma(n) = B(m,n)\Gamma(m+n)$.
For $\int_0^{\pi/2} \sin^{1/2}\theta \cos^{-1/2}\theta\, d\theta$: here $p = 1/2, q = -1/2$, so:

$$I = \frac{\Gamma(3/4)\Gamma(1/4)}{2\Gamma(1)} = \frac{1}{2}\cdot\frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}}$$

🎯 **Verified Result:** $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \frac{\pi}{\sqrt{2}}$.